$$\Rightarrow \frac{15\times 20}{6} = \frac{5\times 12}{x}$$

$$\Rightarrow x = \frac{5\times 12\times 6}{15\times 20} = ₹\ 1.2 \text{ lakh.}$$

Example 4.20 **(Direct – Inverse Variation)**



A mat of length 180 m is made by 15 women in 12 days. How long will it take for 32 women to make a mat of length 512 m?

Solution:

Proportion Method

Length (Work)	Women	Days
180 (*D*) 512	15 (*I*) 32	12 (*D*) (*I*) *x*

Here, the unknown is the days (x). It is to be compared with the length and the women.

Step 1:

Here, more length means more days. So, it is in direct proportion.

$\therefore$ The proportion is $180 : 512 :: 12 : x$ ⟶ (1)

Step 2:

Also, more women means less days. So, it is in inverse proportion.

$\therefore$ The proportion is $32 : 15 :: 12 : x$ ⟶ (2)

Step 3:

Combining (1) and (2),

$$\left.\begin{matrix} 180 : 512 \\ 32 : 15 \end{matrix}\right\} :: 12 : x$$

We know that, **the product of the extremes = the product of the means**

Extremes	:	Means	:	Extremes
180	:	512 : 12	:	*x*
32		15		

So, $180\times 32\times x = 512\times 12\times 15 \Rightarrow x = \frac{512\times 12\times 15}{180\times 32} = 16$ days.

Multiplicative Factor Method

Length (Work)	Women	Days
180 (*D*) 512	15 (*I*) 32	12 (*D*) (*I*) *x*

Here, the unknown is the days (x). It is to be compared with the length and the women.

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